

# **IBM University Engagement Project Submission**

*“Exploring the Power of Agentic AI with IBM Granite and LangFlow ML”*

## Project Tile – **Network Intrusion Detection System (NIDS)**

**Domain of Project – Electronics and Telecommunications Engineering:  
(Machine learning project)**

| Student Name                 | Recent Passport Photo |                                                                                    | Email ID                        | Mobile number [WhatsApp] |
|------------------------------|-----------------------|------------------------------------------------------------------------------------|---------------------------------|--------------------------|
| Harshit Rajkumar Suryawanshi |                       |  | suryawanshiharshit777@gmail.com | 9860745732               |

# Problem Statement - 40

## Problem statement – Network Intrusion Detection (NIDS)

### The Challenge:

**Create a robust network intrusion detection system (NIDS) using machine learning. The system should be capable of analyzing network traffic data to identify and classify various types of cyber-attacks (e.g., DoS, Probe, R2L, U2R) and distinguish them from normal network activity. The goal is to build a model that can effectively secure communication networks by providing an early warning of malicious activities.**

### Objectives of the Network Intrusion Detection System (NIDS)

To develop a machine learning-based Network Intrusion Detection System (NIDS) that can monitor network traffic, detect malicious activities, classify different types of cyber-attacks, and distinguish them from normal network behavior in real time.

### Specific Objectives

- 1. Collect and preprocess network traffic data to remove inconsistencies, handle missing values, and prepare the dataset for machine learning algorithms.**
- 2. Analyze network traffic patterns and identify features that are useful for distinguishing normal traffic from intrusion attempts.**
- 3. Build and train machine learning models capable of detecting network intrusions with high accuracy.**
- 4. Classify different categories of cyber-attacks, including Denial of Service (DoS) , Probe Attacks , Remote-to-Local (R2L) Attack**
- 5. Visualize detection results and attack statistics through dashboards or reports for easier monitoring and analysis.**
- 6. Provide early warning and alert generation when suspicious or malicious network activities are detect**

# Technology Used

## Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models
- Machine-learning tool

I used Machine-Learning-Tool+ IBM watsonx.ai

# Proposed Solution -

## Proposed Solution: Network Intrusion Detection (NIDS)

The proposed solution is to develop a Machine Learning-based Network Intrusion Detection System (NIDS) using IBM Watsonx.ai and the Kaggle Network Intrusion Detection Dataset. The system will analyze network traffic data, preprocess and extract important features, and train a machine learning model to identify suspicious activities. It will classify network traffic into Normal, DoS, Probe, R2L, and U2R categories. The trained model will continuously monitor network traffic and generate real-time alerts when malicious behavior is detected. By using IBM Watsonx.ai, the solution aims to provide accurate, efficient, and automated cyber-attack detection, helping organizations improve their network security and protect against potential threats.

# Machine-Learning component Used -

1) **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the **Network Intrusion Detection (NIDS)** tool. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the network detection tool

2) **IBM watsonx AI Agent** - Processes user input to predict the **Network Intrusion Detection** using given data set

3) **File component** - Stores and retrieves user data , protocol , services , flags , etc for network detection datasets for analysis and tracking.

4) **Pipeline component** – for the deployment of the machine learning tool pipeline component is used

# Machine-Learning workflow

IBM watsonx

Upgrade ⓘ 🔔

Harshit Suryawanshi's Acco... ▾

Sydney ▾

HS

Projects / Network Intrusion Detection / Network Intrusion Detection

Configure AutoAI experiment

Network Intrusion Detection [↗](#)

Autosaved: 9:27:46 PM

Add data source

Add files such as tabular data (CSV).

Browse

Select from project

Train\_data.csv

Size: 2.74 MB Columns: 42

Configure details

Create a time series analysis?

Enable this option to predict future activity over a specified date/time range. Data must be structured and sequential. [Learn more](#)

Yes

No

What do you want to predict?

Prediction column: ⓘ

class

×

▾

Prediction column: class

CUH remaining: 20 CUH

PREDICTION TYPE

Binary Classification

POSITIVE CLASS

Normal

OPTIMIZED FOR

Accuracy & run time

Experiment settings

⚙

Run experiment ▶

IBM watsonx

Upgrade ⓘ 🔔

Harshit Suryawanshi's Acco... ▾

Sydney ▾

HS

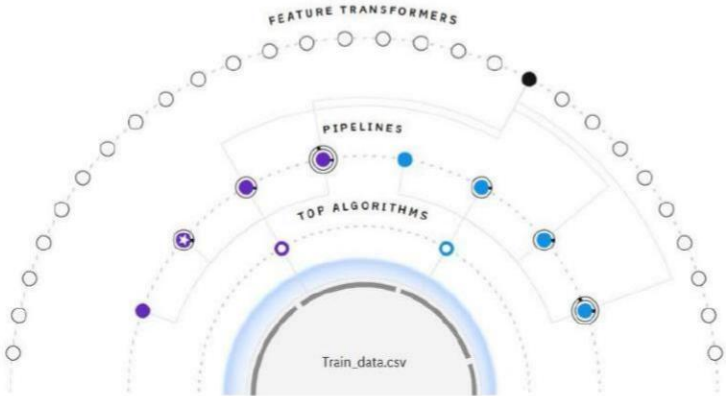
Projects / Network Intrusion Detection / Network Intrusion Detection

Experiment summary Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score [↕](#)

Relationship map ⓘ

Prediction column: class



Progress map

[Swap view](#) [↔](#)



Running

FINISHING UP

Time elapsed: 3 minutes

View log

Save code

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Deployment spaces / ... / P2 - Snap Decision Tree Classifier: Network Intrusion Detection

Deployments

Model details

🔍 Search

New deployment

| Name                                                                                                                        | Type   | Status   | Tags | Last modified                               |
|-----------------------------------------------------------------------------------------------------------------------------|--------|----------|------|---------------------------------------------|
|  Network Intrusion Detection System (NIDS) | Online | Deployed |      | 22 seconds ago<br>Harshit Suryawanshi (you) |

Items per page: 20 1-1 of 1 items 1 of 1 pages

About this asset

Name

P2 - Snap Decision Tree Classifier: Network Intrusion Detection

Description

No description provided.

Asset Details

Type: wml-hybrid\_0.1  
Model ID: 819e936a-34be-78...  
Software specification:  
hybrid\_0.1  
Hybrid pipeline software specifications:  
autoai-kb\_rt25.1-py3.12

Tags

Add tags to make assets easier to find.

Edit

Source asset details

Last modified  
2 minutes ago by Harshit Suryawanshi  
Created on  
Jun 4, 2026 by Harshit Suryawanshi

IBM watsonx

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Deployment spaces / Network Intrusion Detection system (NIDS) / P2 - Snap Decision Tree Classifier: Network Intr...

Network Intrusion Detection System (NIDS) Deployed Online

API reference

Test

Endpoints for inferencing ⓘ

Private endpoint

https://private.au-syd.ml.cloud.ibm.com/ml/v4/deployments/819e936c-2eff-74ba-9f5a-ca4ff2384642/predictions?version=2021-05-01

Public endpoint

https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/819e936c-2eff-74ba-9f5a-ca4ff2384642/predictions?version=2021-05-01

Learn more about the 2021-05-01 version query parameter

Code snippets

cURL

Java

JavaScript

Python

Scala

# NOTE: you must set \$API\_KEY below using information retrieved from your IBM Cloud account (https://au-syd.dal.cloud.ibm.com/docs/content/ibm/cloud/account-credentials)

export API\_KEY=<your API key>

export IAM\_TOKEN=\$(curl --insecure -X POST --location 'https://iam.cloud.ibm.com/identity/token' --header 'Content-Type: application/x-www-form-urlencoded' --data 'grant\_type=urn:ibm:params:oauth:grant-type-token-service' --data 'apikey=\$API\_KEY')

# TODO: manually define and pass values to be scored below

About this deployment

Name

Network Intrusion Detection System (NIDS)

Description

No description provided.

Deployment Details

Deployment ID: 819e936c-2eff-74...  
Serving name:  
No serving name  
Software specification:  
hybrid\_0.1  
Hybrid pipeline software specifications:  
autoai-kb\_rt25.1-py3.12  
Copies:  
1

Tags

Add tags to make assets easier to find.

Associated asset

P2 - Snap Decision Tree Classifier: Network Intrusion Detection

819e936a-34be-7807-802f-7e4475711b27

Last modified

2 minutes ago

# Network Intrusion Detection System (NIDS) 🟢 Deployed Online

API reference

Test

## Enter input data

Text

JSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

[Download CSV template](#) [Browse local files](#) [Search in space](#) [Clear all](#)

|   | duration (integer) | protocol_type (other) | service (other) | flag (other) | src_bytes (integer) | dst_bytes (integer) | land (integer) | wrong_fragment (integer) | urgent (integer) |
|---|--------------------|-----------------------|-----------------|--------------|---------------------|---------------------|----------------|--------------------------|------------------|
| 1 | 0                  | tcp                   | http            | SF           | 8154                | 0                   | 0              | 0                        | 0                |
| 2 | 0                  | tcp                   | private         | SO           | 0                   | 0                   | 0              | 0                        | 0                |
| 3 | 0                  | tcp                   | private         | REJ          | 9786                | 0                   | 0              | 0                        | 0                |
| 4 | 1082               | udp                   | other           | SEF          | 142                 | 105                 | 0              | 0                        | 0                |
| 5 | 0                  | tcp                   | http            | SO           | 1                   | 0                   | 0              | 0                        | 0                |
| 6 |                    |                       |                 |              |                     |                     |                |                          |                  |
| 7 |                    |                       |                 |              |                     |                     |                |                          |                  |
| 8 |                    |                       |                 |              |                     |                     |                |                          |                  |

5 rows, 41 columns

Predict

## Prediction results

Close

✕

### Prediction percentage



### Confidence level distribution



Display format for prediction results

☒ Table view ☐ JSON view

☐ Show input data ⓘ

|    | Prediction | Confidence |
|----|------------|------------|
| 1  | normal     | 100%       |
| 2  | anomaly    | 100%       |
| 3  | anomaly    | 100%       |
| 4  | normal     | 100%       |
| 5  | normal     | 100%       |
| 6  |            |            |
| 7  |            |            |
| 8  |            |            |
| 9  |            |            |
| 10 |            |            |
| 11 |            |            |
| 12 |            |            |
| 13 |            |            |

Download JSON file

## Resource usage

## Project

General

 Access control

## Environments

### Resource usage

 Services & integrations

## Tools

 Pipeline

### Resource usage

### Usage summary

For this month in this project

 $1_{CUH}$ 

0 Tokens

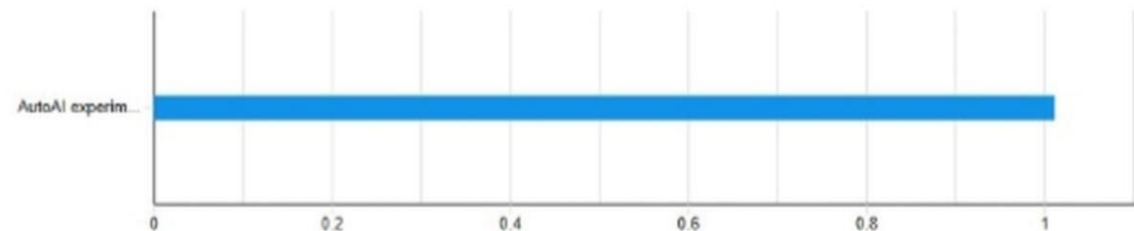
0 Hosting hours

0 Pages

[Manage service instances](#) 

### Usage by type

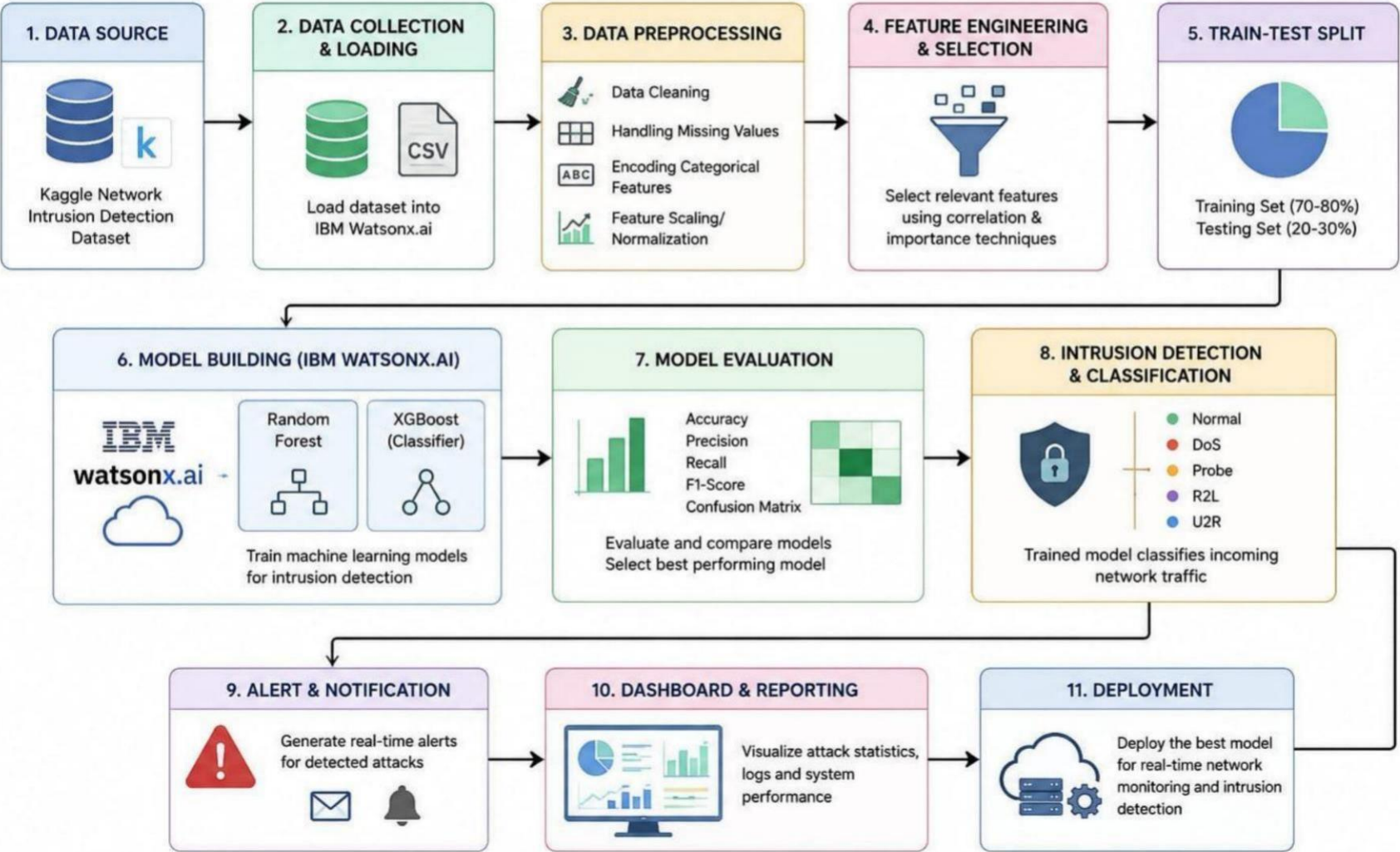
CUH



# Architecture blueprint

## ARCHITECTURE BLUEPRINT

### Network Intrusion Detection System (NIDS)



## Novelty and Uniqueness

- Uses IBM Watsonx.ai to build a cloud-based Machine Learning Network Intrusion Detection System (NIDS).
- Detects both known and unknown cyber threats by learning network traffic patterns.
- Classifies multiple attack types (DoS, Probe, R2L, and U2R) along with normal network traffic.
- Provides real-time intrusion detection and early warning alerts.
- Reduces false positives and false negatives through optimized feature selection and model training.
- Combines automation, scalability, and intelligent threat detection in a single solution.
- Easily adaptable to new and evolving cyber-attack techniques, making it suitable for modern network security environments.

## Git Hub Link

Github URL – [https://github.com/Harshit07-ux-dev/Network\\_Intrusion\\_Detection\\_-NIDS-](https://github.com/Harshit07-ux-dev/Network_Intrusion_Detection_-NIDS-)

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# **Future Scope**

## **Algorithms Used**

- 1. Random Forest**
- 2. XGBoost (Extreme Gradient Boosting)**

## **Future Scope**

- 1. Real-Time Deployment:** Integrate the model with live network monitoring systems to detect and prevent cyber-attacks in real time.
- 2. Advanced AI Integration:** Incorporate Deep Learning and Artificial Intelligence techniques to improve detection of zero-day attacks and emerging cyber threats with higher accuracy.

# Thank You!

Thank you for your time and interest.